

Bats Europe

TRF Binary Order Entry Specification

Version 2.0.17

19 July 2017

Note: Changes highlighted in red are effective from Friday, 27th October 2017

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1 Introduction

1.1 Overview

This document describes Bats TRF Binary Order Entry (TRF BOE), the Bats Europe (hereafter, "Bats") proprietary order entry protocol.

Where applicable, the terminology used in this document is similar to that used by the FIX protocol to allow those familiar with FIX to more easily understand TRF BOE. This document assumes the reader has basic knowledge of the FIX protocol.

TRF BOE fulfills the following requirements:

- *CPU and memory efficiency.* Message encoding, decoding, and parsing are simpler to code and can be optimized to use less CPU and memory at runtime.
- *Application level simplicity.* State transitions are simple and unambiguous. They are easy to apply to a Participant's representation of an order.
- *Session level simplicity.* The session level protocol (login, sequencing, replay of missed messages, logout) is simple to understand.

Whilst Bats has strived to preserve feature parity between FIX and TRF BOE where possible, some features may only be available in one protocol or the other.

All binary values are in little Endian (used by Intel x86 processors), and *not* network byte order.

Each message is identified by a unique message type. Not all message types are used in all of Bats' trading environments globally. A listing of the supported message types is provided in **List of Message Types** (§ 9, p. 62).

All communication is via standard TCP/IP.

1.2 Motivation for Version 2

BOE Version 1 has a number of fixed size parts of messages which, while envisioned to be large enough for future growth, have been unable to accomodate Bats' growth into new service offerings. Version 2 allows greater opportunity for future expansion by eliminating those problems.

Version 2's goals are as follows:

- *Return bitfield expansion.* Messages from Bats to Participant no longer have a limited number of return bitfields. Participants may ignore newly added fields as before, but there is no longer a fixed limit to the number of possible fields returned.
- *Login message parameter groups.* In Version 2, the Login Request V2 message can have extendable parameter groups sent to modify behavior in a forward compatible manner.
- *Easy extension of messages from Participant to Bats to support more bitfields.* In Version 1, messages such as New Order supported a fixed number of bitfields. In Version 2, New Order V2 requires that the number of entered bitfields be specified. This supports, in a backwards compatible way, addition of new bitfields in the future.
- *Easier addition of new messages.* In Version 1, the return bitfields for *all* messages had to be represented in the Login Request. Addition of messages meant changes to the fundamental structure of the Login Request. In Version 2, repeatable parameter groups are used to specify which bitfields are to be sent for different message type. This allows the Login Request V2 to accomodate new message types without fundamental changes to the message structure.

- *Simplification of documentation.* Bats has reduced the complexity of this documentation to make TRF BOE easier to understand.

If you are newly developing to the Bats TRF BOE, you should implement to Version 2 of the specification. Newly added features (e.g., new message fields) *may* be implemented only in Version 2. You may migrate to Version 2 at any point, but you will be *required* to migrate to Version 2 if and when you require use of such features.

To the extent possible, Version 2 has a similar "look and feel" to Version 1. Session-level concepts such as sequencing and heartbeats are identical. Only messages documented in Version 2 are supported on a connection established with a Login Request V2. Data type encoding remains identical. A design goal for the evolution to Version 2 was to make it possible to upgrade Version 1 code to support Version 2 with a minimal amount of development effort.

1.3 Data Types

The following data types are used by TRF BOE. The size of some data types varies by message. All data types have default values of binary zero, in both Participant to Bats and Bats to Participant contexts.

- *Binary*: Little Endian byte order, unsigned binary value. The number of bytes used depends on the context.
  - One byte: `FE` $= 254$
  - Four bytes: `64 00 00 00` $= 100$
- *Signed Binary*: Little Endian byte order, signed two's complement, binary value. The number of bytes used depends on the context.
  - One byte: `DF` $= -33$
  - Four bytes: `64 00 00 00` $= +100$

- *Binary Price*: Little Endian byte order value, signed two's complement, eight bytes in size, with six implied decimal places. So, if the value is $12,340,000$, the actual value taking into account implied decimal places is 12.34.
  - `20 4B BC 00 00 00 00 00` $= 12,340,000/1000000 = 12.34$

  For negative prices, if the value is $-12,340,000$, the actual value taking into account implied decimal places is -12.34.
  - `E0 B4 43 FF FF FF FF FF` $= -12,340,000/1000000 = -12.34$
- *Short Binary Price*: Little Endian byte order value, signed two's complement, four bytes in size, with six implied decimal places. So, if the value is $12,300$, the actual value taking into account implied decimal places is 1.23.
  - `0C 30 00 00` $= 12,300/10000 = 1.23$
- *Trade Price*: Little Endian byte order value, eight bytes in size, with seven implied decimal places. So, if the value is $123,400,000$, the actual value taking into account implied decimal places is 12.34.
  - `40 EF 5A 07 00 00 00 00` $= 123,400,000/10000000 = 12.34$
- *Signed Binary Fee*: Little Endian byte order value, signed two's complement, eight bytes in size, signed, with five implied decimal places. So, the value $-123,000$ is -1.23 after taking account for the five implied decimal places.
  - `88 1F FE FF FF FF FF FF` $= -123,000/100000 = -1.23$
- *Alpha*: Uppercase letters (A–Z) and lowercase letters (a–z) only. ASCII NUL (0x00) filled on the right, if necessary. The number of bytes used depends on the context.
- *Alphanumeric*: Uppercase letters (A–Z), lowercase letters (a–z) and numbers (0–9) only. ASCII NUL (0x00) filled on the right, if necessary.
- *Text*: Printable ASCII characters only. ASCII NUL (0x00) filled on the right, if necessary.
- *DateTime*: 8 bytes. The date and time, in UTC, represented as nanoseconds past the UNIX epoch (00:00:00 UTC on 1 January 1970). The nanoseconds portion is currently ignored and treated as 0 (i.e. the times are only accurate to microseconds) on input, and will always be set to 0 by Bats in outgoing messages. However, Bats may begin populating the nanoseconds portion at any time without warning.

  For example: 1,294,909,373,757,324,000 = `2011-01-13 09:02:53.757324 UTC`.
- *Date*: Little Endian byte order, unsigned binary value, 4 bytes in size. The YYYYMMDD expressed as an integer.

1.4 Optional Fields and Bitfields

Some messages such as Trade Capture Report V2 and Quote V2 have a number of optional fields. A count and number of bitfields in the message specify which optional fields will be present at the end of the message. If a bit is set, the field will be present. Fields are appended to the end of the message. There is no implicit framing between the optional fields. In order to decode the optional fields, they *must* be appended in a particular order to the end of the message. The fields of the first bitfield are appended first, lowest order bit first. Next, the fields of the next bitfield are appended, lowest order bit first. This continues for all bitfields. While certain *reserved* bits within a defined bitfield are used within another Bats market and will be ignored, bits that are reserved for future expansion must be set to 0 when noted in the bitfield description.

The size, data type, and values for each field are described in **List of Optional Fields** (§ 7, p. 54).

Note that the set of optional fields returned for each Bats to Participant message type is determined at session login (using the LOGIN REQUEST V2 message); hence, the exact size and layout of each message received by the client application can be known in advance. Any requested optional field which is irrelevant in a particular context will still be present in the returned message, but with all bytes set to binary zero (0x00).

Each return message from Bats to Participant indicates the optional fields which are present, even though the Participant indicated during login which optional fields are to be sent. The reason for the inclusion (and duplication) is so that each message can be interpreted on its own, without having to find the corresponding login request or response to know which optional fields are present. So, for example, in a log file, decoding a message requires only that single message.

Example messages are shown with each message type which should help to make this concept clear.

2 Session

2.1 Message Headers

Each message has a ten byte header. The two initial *StartOfMessage* bytes are present to aid in message reassembly for network capture purposes. The *MatchingUnit* field is only populated on sequenced non-session level messages sent from Bats to the Participant. Messages from Participant to Bats and all session level messages must always set this value to 0.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	Message type.
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in TRF BOE correspond to matching units on Multicast PITCH. **For session level traffic, the unit is set to 0.** **For messages from Participant to Bats, the unit must be 0.**
SequenceNumber	6	4	Binary	The sequence number for this message. Messages from Bats to Participant are sequenced distinctly per matching unit. Messages from Participant to Bats are sequenced across all matching units with a single sequence stream. Participant can optionally send a 0 sequence number on all messages from Participant to Bats. Bats highly recommends Participant to send sequence number on all inbound messages.

2.2 Login, Replay and Sequencing

Session level messages, both inbound (Participant to Bats) and outbound (Bats to Participant) are unsequenced.

Inbound (Participant to Bats) application messages are sequenced. Upon reconnection, Bats informs the Participant of the last processed sequence number; the Participant *may* choose to resend any messages with sequence numbers greater than this value. A gap forward in the Participant's incoming sequence number is permitted at any time and is ignored by Bats. Gaps backward in sequence number (including the same sequence number used twice) are never permitted and will always result in a LOGOUT message being sent and the connection being dropped.

Most (but not all) outbound (Bats to Participant) application messages are monotonically sequenced per matching unit. Each message's documentation will indicate whether it is sequenced or unsequenced. While matching units on TRF BOE correspond directly to matching units on Multicast PITCH, sequence numbers do not.

Upon reconnection, a Participant sends the last received sequence number per matching unit in a LOGIN REQUEST V2 message. Bats will respond with any missed messages. However, when the LOGIN REQUEST V2 *NoUnspecifiedUnitReplay* flag is enabled, Bats will exclude messages from unspecified matching units during replay. Bats will send a REPLAY COMPLETE message when replay is finished. If there are no messages to replay, a

Replay Complete message will be sent immediately after a Login Response V2 message. Bats will reject all orders during replay.

Assuming Participant has requested replay messages using a properly formatted Login Request V2 after a disconnect, any unacknowledged orders remaining with the Participant after the Replay Complete message is received should be assumed to be unknown to Bats.

Unsequenced messages will not be included during replay.

A session is identified by the username and session sub-identifier (both supplied by Bats). Only one concurrent connection per username and session sub-identifier is permitted.

If a login is rejected, an appropriate Login Response V2 message will be sent and the connection will be terminated.

2.3 Sequence Reset

A reset sequence operation is not available for Binary Order Entry. However, a Participant can send a Login Request message with *NoUnspecifiedUnitReplay* field enabled, and *NumberOfUnits* field set to zero. Then, upon receiving a Login Response V2 message from Bats, the Participant can use the field *LastReceivedSequenceNumber* as the sequence starting point for sending future messages.

2.4 Heartbeats

Client Heartbeat messages are sent from Participant to Bats and Server Heartbeat messages are sent from Bats to Participant if no other data has been sent in that direction for one second. Like other session level messages, heartbeats from Bats to the Participant do *not* increment the sequence number. If Bats receives no inbound data or heartbeats for five seconds, a Logout message will be sent and the connection will be terminated. **Participants are encouraged to have a one second heartbeat interval and to perform similar connection staleness logic.**

2.5 Logging Out

To gracefully log out of a session, a Logout Request message should be sent by the Participant. Bats will finish sending any queued data for that port and will then respond with its own Logout message and close the connection. After receipt of a Logout Request message, Bats will ignore all other inbound (Participant to Bats) messages except for Client Heartbeat.

3 Session Messages

3.1 Participant to Bats

3.1.1 Login Request V2

A Login Request V2 message must be sent as the first message upon connection.

A number of repeating parameter groups, some of which may be required, are sent at the end of the message. Ordering of parameter groups is not important. New parameter groups may be added in the future with no notice.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be 0xBA 0xBA.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	0x37
MatchingUnit	5	1	Binary	Always 0 for inbound (Participant to Bats) messages.
SequenceNumber	6	4	Binary	Always 0 for session level messages.
SessionSubID	10	4	Alphanumeric	Session Sub ID supplied by Bats.
Username	14	4	Alphanumeric	Username supplied by Bats.
Password	18	10	Alphanumeric	Password supplied by Bats.
NumberOfParam Groups	28	1	Binary	A number, n (possibly 0), of parameter groups to follow.
ParamGroup$_1$				First parameter group.
⋮				
ParamGroup$_n$				Last parameter group.

Unit Sequences Parameter Group

This parameter group includes the last consumed sequence number per matching unit received by the Participant. Bats uses these sequence numbers to determine what outbound (Bats to Participant) traffic, if any, was missed by the Participant. If this parameter group is not sent, it's assumed the Participant has not received any messages (e.g., start of day).

The Participant does *not* need to include a sequence number for a unit if they have never received messages from it. For example, if the Participant has received responses from units 1, 3, and 4, the Login Request V2 message need not include unit 2. If the Participant wishes to send a value for unit 2 anyway, 0 would be the only allowed value.

Only one instance of this parameter group may be included.

Field	Offset	Length	Data Type	Description
ParamGroupLength	0	2	Binary	Number of bytes for the parameter group, including this field.
ParamGroupType	2	1	Binary	0x80
NoUnspecified UnitReplay	3	1	Binary	Flag indicating whether to replay missed outgoing (Bats to Participant) messages for unspecified units. 0x00 = False (Replay Unspecified Units) 0x01 = True (Suppress Unspecified Units Replay)

NumberOfUnits	4	1	Binary	A number, n (possibly 0), of unit/sequence pairs to follow, one per unit from which the Participant has received messages.
$\quad$ *UnitNumber*$_1$		1	Binary	A unit number.
$\quad$ *UnitSequence*$_1$		4	Binary	Last received sequence number for the unit.
$\quad \vdots$				
$\quad$ *UnitNumber*$_n$		1	Binary	A unit number.
$\quad$ *UnitSequence*$_n$		4	Binary	Last received sequence number for the unit.

Return Bitfields Parameter Group

This parameter group, which may be repeated, indicates which attributes of a message will be returned by Bats for the remainder of the session. This allows Participants to tailor the echoed results to the needs of their system without paying for bandwidth or processing they do not need.

Listing of the return bitfields which are permitted per message is contained in **Return Bitfields Per Message** (§ 6, p. 35).

Field	Offset	Length	Data Type	Description
ParamGroupLength	0	2	Binary	Number of bytes for the parameter group, including this field.
ParamGroupType	2	1	Binary	0x81
MessageType	3	1	Binary	Return message type for which the bitfields are being specified (e.g., 0x30 for a Trade Capture Report Acknowledgment V2 message)
NumberOfReturn Bitfields	4	1	Binary	Number of bitfields to follow.
ReturnBitfield$_1$	5	1	Binary	Bitfield identifying fields to return.
$\quad \vdots$				
ReturnBitfield$_n$		1	Binary	Last bitfield.

Example Login Request V2 Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	43 00	67 bytes
MessageType	37	Login Request V2
MatchingUnit	00	Always 0 for inbound messages
SequenceNumber	00 00 00 00	Always 0 for session level messages
SessionSubID	30 30 30 31	0001
Username	54 45 53 54	TEST
Password	54 45 53 54 49 4E 47 00 00 00	TESTING
NumberOfParam Groups	03	3 parameter groups
ParamGroupLength	14 00	20 bytes for this parameter group
ParamGroupType	80	0x80 = Unit Sequences
NoUnspecified UnitReplay	01	True (replay only specified units)
NumberOfUnits	03	Three unit/sequence pairs to follow;
$UnitNumber_1$	01	Unit 1
$UnitSequence_1$	4A BB 01 00	Last received sequence of 113,482
$UnitNumber_2$	02	Unit 2
$UnitSequence_2$	00 00 00 00	Last received sequence of 0
$UnitNumber_3$	04	Unit 4
$UnitSequence_3$	79 A1 00 00	Last received sequence of 41,337
ParamGroupLength	08 00	8 bytes for this parameter group
ParamGroupType	81	0x81 = Return Bitfields
MessageType	30	0x30 = Trade Capture Report Acknowledgment V2
NumberOfReturn Bitfields	03	3 bitfields to follow
$ReturnBitfield_1$	00	No bitfields from byte 1
$ReturnBitfield_2$	41	*Symbol*, *Capacity*
$ReturnBitfield_3$	05	*Account*, *ClearingAccount*
ParamGroupLength	0C 00	12 bytes for this parameter group
ParamGroupType	81	0x81 = Return Bitfields
MessageType	32	0x32 = Trade Capture Confirm V2
NumberOfReturn Bitfields	07	7 bitfields to follow
$ReturnBitfield_1$	00	No bitfields from byte 1
$ReturnBitfield_2$	41	*Symbol*, *Capacity*
$ReturnBitfield_3$	07	*Account*, *ClearingFirm*, *ClearingAccount*
$ReturnBitfield_4$	00	No bitfields from byte 4
$ReturnBitfield_5$	00	No bitfields form byte 5
$ReturnBitfield_6$	00	No bitfields from byte 6
$ReturnBitfield_7$	08	*Text*

3.1.2 Logout Request

To end the session, the Participant should send a LOGOUT REQUEST message. Bats will finish sending any queued data and finally respond with a LOGOUT message and close the connection.

A Participant may simply close the connection without logging out, but may lose any queued messages by doing so.

LOGOUT REQUEST remains unchanged between Versions 1 and 2.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be 0xBA 0xBA.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	0x02
MatchingUnit	5	1	Binary	Always 0 for inbound (Participant to Bats) messages.
SequenceNumber	6	4	Binary	Always 0 for session level messages.

Example Logout Request Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	08 00	8 bytes
MessageType	02	Logout Request
MatchingUnit	00	Always 0 for inbound messages
SequenceNumber	00 00 00 00	Always 0 for session level messages

3.1.3 Client Heartbeat

See **Heartbeats** (§ 2.4, p. 9) for more information about heartbeats and the session level protocol.

Client Heartbeat remains unchanged between Versions 1 and 2.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be 0xBA 0xBA.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	0x03
MatchingUnit	5	1	Binary	Always 0 for inbound (Participant to Bats) messages.
SequenceNumber	6	4	Binary	Always 0 for session level messages.

Example Client Heartbeat Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	08 00	8 bytes
MessageType	03	Client Heartbeat
MatchingUnit	00	Always 0 for inbound messages
SequenceNumber	00 00 00 00	Always 0 for session level messages

3.2 Bats to Participant

3.2.1 Login Response V2

A Login Response V2 message is sent in response to a Login Request V2 message. On a successful login, the *LoginResponseStatus* will be set to `A`. On a failed login, *LoginResponseStatus* will be set to a value other than `A`, and *LoginResponseText* will be set to an appropriate failure description.

Bats will verify Return Bitfields at login time. If the Return Bitfields in a Return Bitfields Parameter Group are invalid, *LoginResponseStatus* will be set to `F`, and *LoginResponseText* will include a description of which byte and bit are invalid. This is done to ensure that reserved fields are not used, and only options that apply to the local market are set. See **Return Bitfields Per Message** (§ 6, p. 35) for additional information.

Note that two sets of sequence numbers are available on the Login Response V2. The set of sequence numbers in the body are the actual Bats to Participant sequence numbers indicating the highest sequence numbers available per matching unit. If specified during login, the Unit Sequences Parameter Group will also be returned which is an echo of the sequence numbers the Participant presented during login as the highest received. If these are different, it indicates a gap which will be filled by Bats.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x24`
MatchingUnit	5	1	Binary	Always 0 for session level messages.
SequenceNumber	6	4	Binary	Always 0 for session level messages.
LoginResponse Status	10	1	Alphanumeric	Accepted, or the reason for the rejection. `A` = Login Accepted `N` = Not authorized (invalid username/password) `D` = Session is disabled `B` = Session in use `S` = Invalid session `Q` = Sequence ahead in Login message `I` = Invalid unit given in Login message `F` = Invalid return bitfield in login message `M` = Invalid Login Request message structure
LoginResponse Text	11	60	Text	Human-readable text with additional information about the reason for rejection. For successful logins, this is empty. ASCII NUL (0x00) filled on the right, if necessary.
NoUnspecified UnitReplay	71	1	Binary	Echoed back from the original Login Request V2 message.
LastReceived SequenceNumber	72	4	Binary	Last inbound (Participant to Bats) message sequence number processed by Bats.
NumberOfUnits	76	1	Binary	A number, n, of unit/sequence pairs to follow, one per unit. A pair for every unit will be sent, even if no messages have been sent to this port today. For unsuccessful logins, this will be 0.
UnitNumber$_1$		1	Binary	A unit number.
UnitSequence$_1$		4	Binary	Highest available Bats to Participant sequence number for the unit.
⋮				

UnitNumber$_n$		1	Binary	A unit number.
UnitSequence$_n$		4	Binary	Highest available Bats to Participant sequence number for the unit.
NumberOfParam Groups		1	Binary	Echoed back from the original LOGIN REQUEST V2 message.
ParamGroup$_1$				Echoed back from the original LOGIN REQUEST V2 message.
⋮				
ParamGroup$_n$				Echoed back from the original LOGIN REQUEST V2 message.

Example Login Response V2 Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	88 00	136 bytes
MessageType	24	Login Response V2
MatchingUnit	00	Always 0 for session messages
SequenceNumber	00 00 00 00	Always 0 for session level messages
LoginResponseStatus	41	A = Login Accepted
LoginResponseText	41 63 63 65 70 74 65 64 00 00	Accepted
	00 00 00 00 00 00 00 00 00 00	(padding)
	00 00 00 00 00 00 00 00 00 00	(padding)
	00 00 00 00 00 00 00 00 00 00	(padding)
	00 00 00 00 00 00 00 00 00 00	(padding)
	00 00 00 00 00 00 00 00 00 00	(padding)
NoUnspecified UnitReplay	01	True (replay only specified units)
Last Received Sequence Number	54 4A 02 00	Last sequence Bats received of 150,100
NumberOfUnits	04	Four unit/sequence pairs to follow.
UnitNumber$_1$	01	Unit 1
UnitSequence$_1$	4A BB 01 00	Actual last sequence of 113,482
UnitNumber$_2$	02	Unit 2
UnitSequence$_2$	00 00 00 00	Actual last sequence of 0
UnitNumber$_3$	03	Unit 3
UnitSequence$_3$	00 00 00 00	Actual last sequence of 0
UnitNumber$_4$	04	Unit 4
UnitSequence$_4$	79 A1 00 00	Actual last sequence of 41,337
NumberOfParam Groups	03	3 parameter groups
ParamGroupLength	14 00	20 bytes for this parameter group
ParamGroupType	80	0x80 = Unit Sequences
NoUnspecified UnitReplay	01	True (replay unspecified units)
NumberOfUnits	03	Three unit/sequence pairs to follow
UnitNumber$_1$	01	Unit 1
UnitSequence$_1$	4A BB 01 00	Last received sequence of 113,482
UnitNumber$_2$	02	Unit 2
UnitSequence$_2$	00 00 00 00	Last received sequence of 0
UnitNumber$_3$	04	Unit 4
UnitSequence$_3$	79 A1 00 00	Last received sequence of 41,337
ParamGroupLength	08 00	8 bytes for this parameter group

ParamGroupType	81	0x81 = Return Bitfields
MessageType	30	0x30 = Trade Capture Report Acknowledgment V2
NumberOfReturn Bitfields	03	3 bitfields to follow
$ReturnBitfield_1$	00	No bitfields from byte 1
$ReturnBitfield_2$	41	*Symbol*, *Capacity*
$ReturnBitfield_3$	05	*Account*, *ClearingAccount*
ParamGroupLength	0C 00	12 bytes for this parameter group
ParamGroupType	81	0x81 = Return Bitfields
MessageType	32	0x32 = Trade Capture Confirm V2
NumberOfReturn Bitfields	07	7 bitfields to follow
$ReturnBitfield_1$	00	No bitfields from byte 1
$ReturnBitfield_2$	41	*Symbol*, *Capacity*
$ReturnBitfield_3$	07	*Account*, *ClearingFirm*, *ClearingAccount*
$ReturnBitfield_4$	00	No bitfields from byte 4
$ReturnBitfield_5$	00	No bitfields form byte 5
$ReturnBitfield_6$	00	No bitfields from byte 6
$ReturnBitfield_7$	08	*Text*

3.2.2 Logout

A LOGOUT is usually sent in response to a LOGOUT REQUEST. Any queued data is transmitted, a LOGOUT is sent, and Bats will close the connection. However, a LOGOUT may also be sent if the Participant violates the protocol specification (e.g., by moving backwards in sequence number).

The LOGOUT contains the last transmitted sequence number for each unit, allowing the Participant to check that their last received sequence number matches.

LOGOUT remains unchanged between Versions 1 and 2.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be 0xBA 0xBA.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	0x08
MatchingUnit	5	1	Binary	Always 0 for session level messages.
SequenceNumber	6	4	Binary	Always 0 for session level messages.
LogoutReason	10	1	Alphanumeric	The reason why the LOGOUT message was sent. U = User Requested E = End of Day A = Administrative ! = Protocol Violation
LogoutReason Text	11	60	Text	Human-readable text with additional information about the reason for logout. Particularly useful if *LogoutReason* = ! (Protocol Violation).
LastReceived SequenceNumber	71	4	Binary	Last inbound (Participant to Bats) message sequence number processed by Bats.
NumberOfUnits	75	1	Binary	A number, n (possibly 0), of unit/sequence pairs to follow, one per unit from which the client has received messages.

UnitNumber$_1$		1	Binary	A unit number.
UnitSequence$_1$		4	Binary	Highest available sequence number for the unit.
⋮				
UnitNumber$_n$		1	Binary	A unit number.
UnitSequence$_n$		4	Binary	Highest available sequence number for the unit.

Example Logout Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	55 00	85 bytes
MessageType	08	Logout
MatchingUnit	00	Always 0 for session level messages
SequenceNumber	00 00 00 00	Always 0 for session level messages
LogoutReason	55	U = User Requested
LogoutReason	55 73 65 72 00 00 00 00 00 00	User
Text	00 00 00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00 00 00	
LastReceived SequenceNumber	54 5A 02 00	Last Bats received sequence of 150,100
NumberOfUnits	03	Three unit/sequence pairs to follow.
UnitNumber$_1$	01	Unit 1
UnitSequence$_1$	4A BB 01 00	Last sent sequence of 113,482
UnitNumber$_2$	02	Unit 2
UnitSequence$_2$	00 00 00 00	Last sent sequence of 0
UnitNumber$_3$	04	Unit 4
UnitSequence$_3$	79 A1 00 00	Last sent sequence of 41,337

3.2.3 Server Heartbeat

See **Heartbeats** (§ 2.4, p. 9) for more information about heartbeats and the session level protocol.

SERVER HEARTBEAT remains unchanged between Versions 1 and 2.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be 0xBA 0xBA.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	0x09
MatchingUnit	5	1	Binary	Always 0 for session level messages.
SequenceNumber	6	4	Binary	Always 0 for session level messages.

Example Server Heartbeat Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	08 00	8 bytes
MessageType	09	Server Heartbeat
MatchingUnit	00	Always 0 for session level messages

SequenceNumber	00 00 00 00	Always 0 for session level messages

3.2.4 Replay Complete

See **Login, Replay and Sequencing** (§ 2.2, p. 8) for more information on Login, sequencing and replay. REPLAY COMPLETE remains unchanged between Versions 1 and 2.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be 0xBA 0xBA.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	0x13
MatchingUnit	5	1	Binary	Always 0 for session level messages.
SequenceNumber	6	4	Binary	Always 0 for session level messages.

Example Replay Complete Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	08 00	8 bytes
MessageType	13	Replay Complete
MatchingUnit	00	Always 0 for session level messages
SequenceNumber	00 00 00 00	Always 0 for session level messages

4 Application Messages

4.1 Participant to Bats

4.1.1 Trade Capture Report V2

The TRADE CAPTURE REPORT V2 is used to submit a Trade.

The model supported is as described in the FIX 5.0 (SP2) specification in the Two-Party Reporting workflow diagram of the Trade Capture Reporting section.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x3C`
MatchingUnit	5	1	Binary	Always 0 for inbound (Participant to Bats) messages.
SequenceNumber	6	4	Binary	The sequence number for this message.
TradeReportID	10	20	Text	Corresponds to *TradeReportID* (571) in Bats FIX. Day-unique ID chosen by client. 20 characters or less. Characters in ASCII range 33–126 are allowed, except for comma, semicolon, and pipe. If the *TradeReportID* matches a live trade report (one that has been acked, but not confirmed or declined), it will be rejected as duplicate. **Note: Bats only enforces the uniqueness of TradeReportID values among currently live trade reports. However, we strongly recommend that you keep your TradeReportID values day unique.**
LastShares	30	4	Binary	Corresponds to *LastShares* (32) in Bats FIX. Executed share quantity. If the LargeSize optional field is specified, that value holds precedance over this field.
LastPx	34	8	Trade Price	Corresponds to *LastPx* (31) in Bats FIX. Price of this fill. A value of zero or an indicative price may be used if the price is pending, as denoted by *PriceFormation*
NumberOf TradeCapture ReportBitfields	42	1	Binary	Bitfield identifying bitfields which are set. Field values must be appended to the end of the message.
TradeCapture ReportBitfield$_1$		1	Binary	Bitfield identifying fields to follow.
⋮				
TradeCapture ReportBitfield$_n$		1	Binary	Last bitfield.

NoSides		1	Binary	Corresponds to *NoSides* (552) in Bats FIX. Indicates the number of repeating groups to follow. Currently, can be 1 or 2.

Repeating Group *TrdCapRptSideGrp* must occur the number of times specified in *NoSides*. Only *Side* and *PartyID* are mandatory. Each field occurs in each group, in order as shown below. Optional fields should occur only if corresponding bits in bitfields are set.

Side	1	Alphanumeric	Corresponds to *Side* (54) in Bats FIX. 1 = Buy 2 = Sell 8 = Cross
Capacity	1	Alpha	Corresponds to *OrderCapacity* (47) in Bats FIX. (Orders). Corresponds to *LastCapacity* (29) in Bats FIX. (Executions). A = Agency (maps to 'AOTC') P = Principal (maps to 'DEAL') R = Riskless Principal (maps to 'MTCH')
PartyID	4	Alpha	Corresponds to *PartyID* (448) in Bats FIX. The end-client responsible for the trade. Must be an identifier (4 uppercase letters) known to Bats.
Account	16	Text	Corresponds to *Account* (1) in Bats FIX. Contains the *Account* specified on this leg on the trade capture, if any. Reflected back on trade capture report confirmations. Allowed characters are alphanumeric and colon.
PartyRole	1	Alphanumeric	Corresponds to *PartyRole* (452) in Bats FIX. Contains the *PartyRole* specified on this leg on the trade capture, if any. Reflected back on trade capture report confirmations. 2 = EnteringFirm (the party reporting the trade)

Optional fields...				Optional fields as set in the bitmap. Note, optional fields that occur in the repeating groups appear above, repeating per group, not within this block.

Example Trade Capture Report V2 Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes
MessageLength	4D 00	77 bytes
MessageType	3C	Trade Capture Report V2
MatchingUnit	00	Always 0 for inbound messages
SequenceNumber	64 00 00 00	Sequence number 100
TradeReportID	31 34 32 39 30 39 38 34 38 39 35 38 37 33 33 32 00 00 00 00	1429098489587332
LastShares	46 00 00 00	70 shares
LastPx	40 F9 A1 6A 00 00 00 00	$1789000000 = 178.9000000$
NumberOf TradeCaptureReport Bitfields	04	4 bitfields to follow
$Bitfield_1$	01	*Symbol*
$Bitfield_2$	B5	*Capacity*, *TransactionCategory*, *PartyRole*, *TradeReportTransType*, *VenueType*
$Bitfield_3$	A2	*MatchType*, *TradePublishIndicator*, *ExecutionMethod*
$Bitfield_4$	43	*TradeReportType*, *TradeHandlingInstruction*, *OrderCategory*
NoSides	02	2 repeating groups to follow
Side	31	Buy
Capacity	50	Principal
PartyID	54 45 53 54	TEST
PartyRole	31	ExecutingFirm
Side	32	Sell
Capacity	50	Principal
PartyID	54 45 53 54	TEST
PartyRole	31	ExecutingFirm
Symbol	56 4F 44 6C 00 00 00 00	VOD1
TransactionCategory	50	P = Regular Trade
TradeReportTransType	00	0 = New
VenueType	4F	O = Off Book
MatchType	03	3 = Trade Reporting (On-Exchange)
TradePublishIndicator	01	1 = Publish trade
ExecutionMethod	55	U = Unspecified
TradeReportType	00	0 = Submit
TradeHandlingInstr	01	1 = Two-Party Report
OrderCategory	03	3 = Privately Negotiated Trade

4.1.2 Quote V2

The QUOTE V2 is used to submit a Quote.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be 0xBA 0xBA.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	0x3D

MatchingUnit	5	1	Binary	Always 0 for inbound (Participant to Bats) messages.
SequenceNumber	6	4	Binary	The sequence number for this message.
QuoteID	10	20	Text	Corresponds to *QuoteID* (117) in Bats FIX. Mandatory unique ID chosen by client for a Quote or QuoteCancel. 18 characters or less. Characters in ASCII range 33–126 are allowed, except for comma, semicolon, and pipe. Please note that the last two bytes are for internal use. **Note: Bats only enforces the uniqueness of QuoteID values among currently live quotes. However, we strongly recommend that you keep your QuoteID values day unique.**
NumberOf Quote Bitfields	30	1	Binary	Bitfield identifying bitfields which are set. Field values must be appended to the end of the message.
Quote Bitfield$_1$	31	1	Binary	Bitfield identifying fields to follow.
⋮				
Quote Bitfield$_n$		1	Binary	Last bitfield.
Optional fields...				

4.1.3 Quote Cancel V2

The QUOTE CANCEL V2 is used to cancel a Quote.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x3E`
MatchingUnit	5	1	Binary	Always 0 for inbound (Participant to Bats) messages.
SequenceNumber	6	4	Binary	The sequence number for this message.
QuoteID	10	20	Text	Corresponds to *QuoteID* (117) in Bats FIX. Mandatory unique ID chosen by client for a Quote or QuoteCancel. 18 characters or less. Characters in ASCII range 33–126 are allowed, except for comma, semicolon, and pipe. Please note that the last two bytes are for internal use. **Note: Bats only enforces the uniqueness of QuoteID values among currently live quotes. However, we strongly recommend that you keep your QuoteID values day unique.**
NumberOf QuoteCancel Bitfields	30	1	Binary	Bitfield identifying bitfields which are set. Field values must be appended to the end of the message.
QuoteCancel Bitfield$_1$	31	1	Binary	Bitfield identifying fields to follow.
⋮				
QuoteCancel Bitfield$_n$		1	Binary	Last bitfield.
NoQuoteEntries		1	Binary	Mandatory. Corresponds to *NoQuoteEntriesText* (295) in Bats FIX. Indicates the number of repeating groups to follow. Currently, can be 0 or 1. 0 - Cancel all symbols 1 - Cancel quote for the specified symbol

Repeating Group *QuoteCancelRptGrp* must occur the number of times specified in *NoQuoteEntries*. All fields are optional. Each field occurs in each group, in bitfield order as shown below, if its corresponding bit in the bitfields bit is set.

Symbol	8	Alphanumeric	Corresponds to *Symbol* (55) in Bats FIX. Uniform symbology identifier for the instrument.
Currency	3	Alpha	Corresponds to *Currency* (15) in Bats FIX. ISO currency. Required if *IDSource* is set to 4 (ISIN).
IDSource	1	Alphanumeric	Corresponds to *IDSource* (22) in Bats FIX. 4 = ISIN 5 = RIC
SecurityID	16	Text	Corresponds to *SecurityID* (48) in Bats FIX. ISIN, or RIC if *IDSource* is set.
SecurityExchange	4	Alphanumeric	Corresponds to *SecurityExchange* (207) in Bats FIX. Required if *IDSource* is set to 4 (ISIN).

Optional fields...				Optional fields as set in the bitmap. Note, optional fields that occur in the repeating groups appear above, repeating per group, not within this block. If *ClearingFirm* is not specified, default firm will be used if possible.

4.2 Bats to Participant

4.2.1 Trade Capture Report Acknowledgment V2

The TRADE CAPTURE REPORT ACKNOWLEDGMENT V2 is sent by Bats to acknowledge the receipt of a TRADE CAPTURE REPORT V2. It is a technical-level ack. The Trade is not considered to have fully succeeded until a TRADE CAPTURE CONFIRM V2 is sent.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	0x30
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in TRF BOE correspond to matching units on Multicast PITCH.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per matching unit.
TransactionTime	10	8	DateTime	The time the event occurred in the Bats matching engine (not the time the message was sent).
TradeReportID	18	20	Text	Corresponds to *TradeReportID* (571) in Bats FIX. Contains the *TradeReportID* (571) of the original trade capture report to which this message relates
ReservedInternal	38	1	Binary	Reserved for Bats' internal use.
NumberOfReturn Bitfields	39	1	Binary	Number of bitfields to follow.
$ReturnBitfield_1$	40	1	Binary	Bitfield identifying fields to return.
⋮				
$ReturnBitfield_n$		1	Binary	Last bitfield.
NoSides		1	Binary	Corresponds to *NoSides* (552) in Bats FIX. Indicates the number of repeating groups to follow. Currently, can be 1 or 2.

Repeating Group *TrdCapAckSideGrp* must occur the number of times specified in *NoSides*. All fields are optional. Each field occurs in each group, in bitfield order as shown below, if its corresponding bit in the bitfields bit is set.

Side	1	Alphanumeric	Echoed back from the original TRADE CAPTURE REPORT V2 message.
Capacity	1	Alpha	Echoed back from the original TRADE CAPTURE REPORT V2 message.
Account	16	Text	Echoed back from the original TRADE CAPTURE REPORT V2 message.
PartyID	4	Alpha	Echoed back from the original TRADE CAPTURE REPORT V2 message.
PartyRole	1	Alphanumeric	Echoed back from the original TRADE CAPTURE REPORT V2 message.

Optional fields...				Optional fields as set in the bitmap. Note, optional fields that occur in the repeating groups appear above, repeating per group, not within this block.

4.2.2 Trade Capture Report Reject V2

The Trade Capture Report Reject V2 is sent by Bats in response to a Trade Capture Report V2. Trade Capture Report Reject V2 messages are unsequenced.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x31`
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in TRF BOE correspond to matching units on Multicast PITCH.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per matching unit.
TransactionTime	10	8	DateTime	The time the event occurred in the Bats matching engine (not the time the message was sent).
TradeReportID	18	20	Text	Corresponds to *TradeReportID* (571) in Bats FIX. Contains the *TradeReportID* (571) of the original trade capture report to which this message relates
Reason	38	1	Text	Reason for a Trade Capture Report reject or decline. See **Reason Codes** (§ 8, p. 61) for a list of possible reasons.
Text	39	60	Text	Human readable text with more information about the reject reason.
ReservedInternal	99	1	Binary	Reserved for Bats' internal use.
NumberOfReturn Bitfields	100	1	Binary	Number of bitfields to follow.
$ReturnBitfield_1$	101	1	Binary	Bitfield identifying fields to return.
⋮				
$ReturnBitfield_n$		1	Binary	Last bitfield.
NoSides		1	Binary	Corresponds to *NoSides* (552) in Bats FIX. Indicates the number of repeating groups to follow. Currently, can be 1 or 2.

Repeating Group *TrdCapAckSideGrp* must occur the number of times specified in *NoSides*. All fields are optional. Each field occurs in each group, in bitfield order as shown below, if its corresponding bit in the bitfields bit is set.

Side	1	Alphanumeric	Echoed back from the original TRADE CAPTURE REPORT V2 message.
Capacity	1	Alpha	Echoed back from the original TRADE CAPTURE REPORT V2 message.
Account	16	Text	Echoed back from the original TRADE CAPTURE REPORT V2 message.
PartyID	4	Alpha	Echoed back from the original TRADE CAPTURE REPORT V2 message.
PartyRole	1	Alphanumeric	Echoed back from the original TRADE CAPTURE REPORT V2 message.

Optional fields...				Optional fields as set in the bitmap. Note, optional fields that occur in the repeating groups appear above, repeating per group, not within this block.

4.2.3 Trade Capture Confirm V2

The TRADE CAPTURE CONFIRM V2 is sent from Bats to the participant in order to confirm that a TRADE CAPTURE REPORT V2 has been fully processed. It is a business-level confirmation as distinct from the technology level acknowledgment sent as a TRADE CAPTURE REPORT ACKNOWLEDGMENT V2.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x32`
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in TRF BOE correspond to matching units on Multicast PITCH.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per matching unit.
TransactionTime	10	8	DateTime	The time the event occurred in the Bats matching engine (not the time the message was sent).
TradeReportID	18	20	Text	Corresponds to *TradeReportID* (571) in Bats FIX. Unique identifier for the trade report confirm as provided by Bats
TradeReportRefID	38	20	Text	Corresponds to *TradeReportRefID* (572) in Bats FIX. Contains the *TradeReportID* (571) of the original trade capture report to which this message relates

TradeID	58	8	Binary	Corresponds to *TradeID* (1003) in FIX. An ID allocated by Bats in response to a trade capture report, identifying a particular trade. These are present in the PITCH Off-Book Trade messages, and are guaranteed unique for a minimum of 7 calendar days from the original report.
LastShares	66	4	Binary	Corresponds to *LastShares* (32) in Bats FIX. Executed share quantity. If the LargeSize optional field is specified, that value holds precedance over this field.
LastPx	70	8	Trade Price	Corresponds to *LastPx* (31) in Bats FIX. Price of this fill. A value of zero or an indicative price may be used if the price is pending, as denoted by *PriceFormation*
ContraBroker	78	4	Alphanumeric	Corresponds to *ContraBroker* (375) in Bats FIX. Indicates the market of execution. [1]
ReservedInternal	82	1	Binary	Reserved for Bats' internal use.
NumberOfReturn Bitfields	83	1	Binary	Number of bitfields to follow.
$ReturnBitfield_1$	84	1	Binary	Bitfield identifying fields to return.
⋮				
$ReturnBitfield_n$		1	Binary	Last bitfield.
NoSides		1	Binary	Corresponds to *NoSides* (552) in Bats FIX. Indicates the number of repeating groups to follow. Currently, can be 1 or 2.

Repeating Group *TrdCapAckSideGrp* must occur the number of times specified in *NoSides*. All fields are optional. Each field occurs in each group, in bitfield order as shown below, if its corresponding bit in the bitfields bit is set.

Side	1	Alphanumeric	Echoed back from the original TRADE CAPTURE REPORT V2 message.
Capacity	1	Alpha	Echoed back from the original TRADE CAPTURE REPORT V2 message.
Account	16	Text	Echoed back from the original TRADE CAPTURE REPORT V2 message.
PartyID	4	Alpha	Echoed back from the original TRADE CAPTURE REPORT V2 message.
Central Counterparty	1	Alpha	The CCP handling the trade N = None
PartyRole	1	Alphanumeric	Echoed back from the original TRADE CAPTURE REPORT V2 message.
FeeCode	2	Alphanumeric	Indicates fee associated with an execution. Fee codes are published in the pricing schedule. New fee codes may be sent with little to no notice. Participants are encouraged to code their systems to accept unknown fee codes.

[1] for historical reasons a local execution for an order entered into the BXE book is identified with a value of BATS

Optional fields...				Optional fields as set in the bitmap. Note, optional fields that occur in the repeating groups appear above, repeating per group, not within this block.

4.2.4 Trade Capture Report Decline V2

The Trade Capture Decline V2 is sent from Bats to the participant in order to decline a Trade Capture Report V2. It is a business-level reject as distinct from the technology level acknowledgment sent as a Trade Capture Report Acknowledgment V2.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x33`
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in TRF BOE correspond to matching units on Multicast PITCH.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per matching unit.
TransactionTime	10	8	DateTime	The time the event occurred in the Bats matching engine (not the time the message was sent).
TradeReportID	18	20	Text	Corresponds to *TradeReportID* (571) in Bats FIX. Unique identifier for the trade report confirm as provided by Bats
TradeReportRefID	38	20	Text	Corresponds to *TradeReportRefID* (572) in Bats FIX. Contains the *TradeReportID* (571) of the original trade capture report to which this message relates
TradeID	58	8	Binary	Corresponds to *TradeID* (1003) in FIX. An ID allocated by Bats in response to a trade capture report, identifying a particular trade. These are present in the PITCH Off-Book Trade messages, and are guaranteed unique for a minimum of 7 calendar days from the original report.
LastShares	66	4	Binary	Corresponds to *LastShares* (32) in Bats FIX. Executed share quantity. If the LargeSize optional field is specified, that value holds precedance over this field.
LastPx	70	8	Trade Price	Corresponds to *LastPx* (31) in Bats FIX. Price of this fill. A value of zero or an indicative price may be used if the price is pending, as denoted by *PriceFormation*

ContraBroker	78	4	Alphanumeric	Corresponds to *ContraBroker* (375) in Bats FIX. Indicates the market of execution. [2]
Reason	82	1	Text	Reason for a TRADE CAPTURE REPORT reject or decline. See **Reason Codes** (§ 8, p. 61) for a list of possible reasons.
Text	83	60	Text	Human readable text with more information about the reject reason.
ReservedInternal	143	1	Binary	Reserved for Bats' internal use.
NumberOfReturn Bitfields	144	1	Binary	Number of bitfields to follow.
$ReturnBitfield_1$	145	1	Binary	Bitfield identifying fields to return.
⋮				
$ReturnBitfield_n$		1	Binary	Last bitfield.
NoSides		1	Binary	Corresponds to *NoSides* (552) in Bats FIX. Indicates the number of repeating groups to follow. Currently, can be 1 or 2.
Repeating Group *TrdCapAckSideGrp* must occur the number of times specified in *NoSides*. All fields are optional. Each field occurs in each group, in bitfield order as shown below, if its corresponding bit in the bitfields bit is set.				
Side	1	Alphanumeric	Echoed back from the original TRADE CAPTURE REPORT V2 message.	
Capacity	1	Alpha	Echoed back from the original TRADE CAPTURE REPORT V2 message.	
Account	16	Text	Echoed back from the original TRADE CAPTURE REPORT V2 message.	
PartyID	4	Alpha	Echoed back from the original TRADE CAPTURE REPORT V2 message.	
PartyRole	1	Alphanumeric	Echoed back from the original TRADE CAPTURE REPORT V2 message.	
Optional fields...				Optional fields as set in the bitmap. Note, optional fields that occur in the repeating groups appear above, repeating per group, not within this block.

4.2.5 Quote Status V2

The QUOTE STATUS V2 is sent by Bats in response to a QUOTE V2 or QUOTECANCEL V2. QUOTE STATUS V2 messages are sequenced.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.

[2] for historical reasons a local execution for an order entered into the BXE book is identified with a value of `BATS`

MessageType	4	1	Binary	0x3F
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in TRF BOE correspond to matching units on Multicast PITCH.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per matching unit.
QuoteID	10	20	Text	Corresponds to *QuoteID* (117) in Bats FIX. Mandatory unique ID chosen by client for a Quote or QuoteCancel. 18 characters or less. Characters in ASCII range 33–126 are allowed, except for comma, semicolon, and pipe. Please note that the last two bytes are for internal use. **Note: Bats only enforces the uniqueness of QuoteID values among currently live quotes. However, we strongly recommend that you keep your QuoteID values day unique.**
Status	30	1	Alphanumeric	Indicates the acceptance or otherwise of a Quote or QuoteCancel message. 0 = Accepted in response to a QUOTE message 1 = Cancelled in response to a QUOTECANCEL message
ReservedInternal	31	1	Binary	Reserved for Bats' internal use.
NumberOfReturn Bitfields	32	1	Binary	Number of bitfields to follow.
$ReturnBitfield_1$	33	1	Binary	Bitfield identifying fields to return.
⋮				
$ReturnBitfield_n$		1	Binary	Last bitfield.
Optional fields...				

4.2.6 Quote Reject V2

The QUOTE REJECT V2 is sent by Bats in response to a QUOTE V2 or QUOTECANCEL V2. QUOTE REJECT V2 messages are unsequenced.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be 0xBA 0xBA.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	0x40
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in TRF BOE correspond to matching units on Multicast PITCH.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per matching unit.

QuoteID	10	20	Text	Corresponds to *QuoteID* (117) in Bats FIX. Mandatory unique ID chosen by client for a Quote or QuoteCancel. 18 characters or less. Characters in ASCII range 33–126 are allowed, except for comma, semicolon, and pipe. Please note that the last two bytes are for internal use. **Note: Bats only enforces the uniqueness of QuoteID values among currently live quotes. However, we strongly recommend that you keep your QuoteID values day unique.**
Reason	30	1	Text	Reason for a QUOTE reject. A = Admin Y = Symbol Not Supported Z = Unforseen Reason
Text	31	60	Text	Human readable text with more information about the reject reason.
ReservedInternal	91	1	Binary	Reserved for Bats' internal use.
NumberOfReturn Bitfields	92	1	Binary	Number of bitfields to follow.
$ReturnBitfield_1$	93	1	Binary	Bitfield identifying fields to return.
⋮				
$ReturnBitfield_n$		1	Binary	Last bitfield.
Optional fields...				

5 Input Bitfields Per Message

Legend:

• Indicates that the field can be requested for a message
– Indicates that the field cannot be requested for a message

5.1 Trade Capture Report V2

Byte	Bit	Field	
1	1	*Symbol*	•
	2	*Reserved*	–
	4	*Currency*	•
	8	*IDSource*	•
	16	*SecurityID*	•
	32	*Security Exchange*	•
	64	*ExecInst*	–
	128	*Reserved*	–
2	1	*Capacity*	•
	2	*Account*	•
	4	*TransactionCategory*	•
	8	*TradeTime*	•
	16	*PartyRole*	•
	32	*TradeReportTransType*	•
	64	*TradeID*	•
	128	*VenueType*	•
3	1	*TradingSessionSubId*	•
	2	*MatchType*	•
	4	*TrdSubType*	•
	8	*SecondaryTrdType*	•
	16	*TradePriceCondition*	•
	32	*TradePublishIndicator*	•
	64	*LargeSize*	•
	128	*ExecutionMethod*	•
4	1	*TradeReportType*	–
	2	*TradeHandlingInstruction*	–
	4	*TradeLinkID*	–
	8	*TradeReportRefID*	–
	16	*GrossTradeAmt*	•
	32	*Tolerance*	–
	64	*OrderCategory*	•
	128	*SettlementPrice*	–
5	1	*SettlementDate*	–
	2	*PriceFormation*	•
	4	*AlgorithmicIndicator*	•
	8	*WaiverType*	•
	16	*DeferralReason*	•
	32	*Reserved*	–
	64	*Reserved*	–
	128	*Reserved*	–

5.2 Quote V2

Byte	Bit	Field	
1	1	*Symbol*	•
	2	*Currency*	•
	4	*IDSource*	•
	8	*SecurityID*	•
	16	*SecurityExchange*	•
	32	*Bid*	•
	64	*Offer*	•
	128	*ClearingFirm*	•

5.3 Quote Cancel V2

Byte	Bit	Field	
1	1	*Symbol*	•
	2	*Currency*	•
	4	*IDSource*	•
	8	*SecurityID*	•
	16	*SecurityExchange*	•
	32	*ClearingFirm*	•
	64	*Reserved*	–
	128	*Reserved*	–

6 Return Bitfields Per Message

Legend:

• Indicates that the field can be requested for a message
– Indicates that the field cannot be requested for a message

6.1 Trade Capture Report Acknowledgment V2

Byte	Bit	Field	
1	1	*Side*	•
	2	*PegDifference*	–
	4	*Price*	–
	8	*ExecInst*	•
	16	*OrdType*	–
	32	*TimeInForce*	–
	64	*MinQty*	–
	128	*MaxRemovePct*	–
2	1	*Symbol*	•
	2	*SymbolSfx*	–
	4	*Currency*	•
	8	*IdSource*	•
	16	*SecurityId*	•
	32	*SecurityExchange*	•
	64	*Capacity*	•
	128	*(Reserved)*	–
3	1	*Account*	•
	2	*ClearingFirm*	•
	4	*ClearingAccount*	–
	8	*DisplayIndicator*	–
	16	*MaxFloor*	–
	32	*DiscretionAmount*	–
	64	*OrderQty*	•
	128	*PreventParticipantMatch*	–
4	1	*MaturityDate*	–
	2	*StrikePrice*	–
	4	*PutOrCall*	–
	8	*OpenClose*	–
	16	*ClOrdIdBatch*	–
	32	*CorrectedSize*	–
	64	*PartyID*	•
	128	*AccessFee*	–
5	1	*OrigClOrdId*	–
	2	*LeavesQty*	–
	4	*LastShares*	–
	8	*LastPrice*	–
	16	*DisplayPrice*	–
	32	*WorkingPrice*	–
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	–
6	1	*SecondaryOrderId*	–
	2	*CCP*	–
	4	*ContraCapacity*	–
	8	*AttributedQuote*	–
	16	*ExtExecInst*	–
	32	*BulkOrderIds*	–
	64	*BulkRejectReasons*	–
	128	*PartyRole*	•

continued. . .

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	•
	4	*TradePublishIndReturn*	–
	8	*Text*	–
	16	*Bid*	–
	32	*Offer*	–
	64	*LargeSize*	•
	128	*LastMkt*	–
8	1	*FeeCode*	–
	2	*EchoText*	–
	4	*StopPx*	–
	8	*RoutingInst*	–
	16	*RoutStrategy*	–
	32	*RouteDeliveryMethod*	–
	64	*ExDestination*	–
	128	*TradeReportRefID*	•
9	1	*MarketingFeeCode*	–
	2	*TargetPartyID*	–
	4	*AuctionId*	–
	8	*OrderCategory*	•
	16	*LiquidityProvision*	–
	32	*CmtaNumber*	–
	64	*CrossType*	–
	128	*CrossPrioritization*	–
10	1	*CrossId*	–
	2	*AllocQty*	–
	4	*GiveUpFirmID*	–
	8	*RoutingFirmID*	–
	16	*WaiverType*	•
	32	*CrossExclusionIndicator*	–
	64	*PriceFormation*	•
	128	*ClientQualifiedRole*	–
11	1	*ClientID*	–
	2	*InvestorID*	–
	4	*ExecutorID*	–
	8	*OrderOrigination*	–
	16	*AlgorithmicIndicator*	•
	32	*DeferralReason*	•
	64	*InvestorQualifiedRole*	–
	128	*ExecutorQualifiedRole*	–
12	1	*CtiCode*	–
	2	*ManualOrderIndicator*	–
	4	*OperatorId*	–
	8	*TradeDate*	–
	16	*VariancePrice*	–
	32	*VarianceSize*	–
	64	*OrigSymbolID*	–
	128	*OrigTASPrice*	–

continued. . .

Byte	Bit	Field	
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	–
	128	*MultilegReportingType*	–
14	1	*LegCFICode*	–
	2	*LegMaturityDate*	–
	4	*LegStrikePrice*	–
	8	*(Reserved)*	–
	16	*SecondaryExecId*	–
	32	*(Reserved)*	–
	64	*(Reserved)*	–
	128	*(Reserved)*	–

6.2 Trade Capture Report Reject V2

Byte	Bit	Field	
1	1	*Side*	•
	2	*PegDifference*	–
	4	*Price*	–
	8	*ExecInst*	•
	16	*OrdType*	–
	32	*TimeInForce*	–
	64	*MinQty*	–
	128	*MaxRemovePct*	–
2	1	*Symbol*	•
	2	*SymbolSfx*	–
	4	*Currency*	•
	8	*IdSource*	•
	16	*SecurityId*	•
	32	*SecurityExchange*	•
	64	*Capacity*	•
	128	*(Reserved)*	–
3	1	*Account*	•
	2	*ClearingFirm*	•
	4	*ClearingAccount*	–
	8	*DisplayIndicator*	–
	16	*MaxFloor*	–
	32	*DiscretionAmount*	–
	64	*OrderQty*	•
	128	*PreventParticipantMatch*	–
4	1	*MaturityDate*	–
	2	*StrikePrice*	–
	4	*PutOrCall*	–
	8	*OpenClose*	–
	16	*ClOrdIdBatch*	–
	32	*CorrectedSize*	–
	64	*PartyID*	•
	128	*AccessFee*	–
5	1	*OrigClOrdId*	–
	2	*LeavesQty*	–
	4	*LastShares*	–
	8	*LastPrice*	–
	16	*DisplayPrice*	–
	32	*WorkingPrice*	–
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	–
6	1	*SecondaryOrderId*	–
	2	*CCP*	–
	4	*ContraCapacity*	–
	8	*AttributedQuote*	–
	16	*ExtExecInst*	–
	32	*BulkOrderIds*	–
	64	*BulkRejectReasons*	–
	128	*PartyRole*	•

continued. . .

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	•
	4	*TradePublishIndReturn*	•
	8	*Text*	•
	16	*Bid*	–
	32	*Offer*	–
	64	*LargeSize*	•
	128	*LastMkt*	–
8	1	*FeeCode*	–
	2	*EchoText*	•
	4	*StopPx*	–
	8	*RoutingInst*	–
	16	*RoutStrategy*	–
	32	*RouteDeliveryMethod*	–
	64	*ExDestination*	–
	128	*TradeReportRefID*	–
9	1	*MarketingFeeCode*	–
	2	*TargetPartyID*	–
	4	*AuctionId*	–
	8	*OrderCategory*	–
	16	*LiquidityProvision*	–
	32	*CmtaNumber*	–
	64	*CrossType*	–
	128	*CrossPrioritization*	–
10	1	*CrossId*	–
	2	*AllocQty*	–
	4	*GiveUpFirmID*	–
	8	*RoutingFirmID*	–
	16	*WaiverType*	•
	32	*CrossExclusionIndicator*	–
	64	*PriceFormation*	•
	128	*ClientQualifiedRole*	–
11	1	*ClientID*	–
	2	*InvestorID*	–
	4	*ExecutorID*	–
	8	*OrderOrigination*	–
	16	*AlgorithmicIndicator*	•
	32	*DeferralReason*	•
	64	*InvestorQualifiedRole*	–
	128	*ExecutorQualifiedRole*	–
12	1	*CtiCode*	–
	2	*ManualOrderIndicator*	–
	4	*OperatorId*	–
	8	*TradeDate*	–
	16	*VariancePrice*	–
	32	*VarianceSize*	–
	64	*OrigSymbolID*	–
	128	*OrigTASPrice*	–

continued. . .

Byte	Bit	Field	
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	–
	128	*MultilegReportingType*	–
14	1	*LegCFICode*	–
	2	*LegMaturityDate*	–
	4	*LegStrikePrice*	–
	8	*(Reserved)*	–
	16	*SecondaryExecId*	–
	32	*(Reserved)*	–
	64	*(Reserved)*	–
	128	*(Reserved)*	–

6.3 Trade Capture Confirm V2

Byte	Bit	Field	
1	1	*Side*	•
	2	*PegDifference*	–
	4	*Price*	–
	8	*ExecInst*	•
	16	*OrdType*	–
	32	*TimeInForce*	–
	64	*MinQty*	–
	128	*MaxRemovePct*	–
2	1	*Symbol*	•
	2	*SymbolSfx*	–
	4	*Currency*	•
	8	*IdSource*	•
	16	*SecurityId*	•
	32	*SecurityExchange*	•
	64	*Capacity*	•
	128	*(Reserved)*	–
3	1	*Account*	•
	2	*ClearingFirm*	•
	4	*ClearingAccount*	–
	8	*DisplayIndicator*	–
	16	*MaxFloor*	–
	32	*DiscretionAmount*	–
	64	*OrderQty*	•
	128	*PreventParticipantMatch*	–
4	1	*MaturityDate*	–
	2	*StrikePrice*	–
	4	*PutOrCall*	–
	8	*OpenClose*	–
	16	*ClOrdIdBatch*	–
	32	*CorrectedSize*	•
	64	*PartyID*	•
	128	*AccessFee*	–
5	1	*OrigClOrdId*	–
	2	*LeavesQty*	–
	4	*LastShares*	–
	8	*LastPrice*	–
	16	*DisplayPrice*	–
	32	*WorkingPrice*	–
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	–
6	1	*SecondaryOrderId*	–
	2	*CCP*	•
	4	*ContraCapacity*	–
	8	*AttributedQuote*	–
	16	*ExtExecInst*	–
	32	*BulkOrderIds*	–
	64	*BulkRejectReasons*	–
	128	*PartyRole*	•

continued. . .

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	•
	4	*TradePublishIndReturn*	•
	8	*Text*	•
	16	*Bid*	–
	32	*Offer*	–
	64	*LargeSize*	•
	128	*LastMkt*	–
8	1	*FeeCode*	•
	2	*EchoText*	•
	4	*StopPx*	–
	8	*RoutingInst*	–
	16	*RoutStrategy*	–
	32	*RouteDeliveryMethod*	–
	64	*ExDestination*	–
	128	*TradeReportRefID*	–
9	1	*MarketingFeeCode*	–
	2	*TargetPartyID*	–
	4	*AuctionId*	–
	8	*OrderCategory*	•
	16	*LiquidityProvision*	–
	32	*CmtaNumber*	–
	64	*CrossType*	–
	128	*CrossPrioritization*	–
10	1	*CrossId*	–
	2	*AllocQty*	–
	4	*GiveUpFirmID*	–
	8	*RoutingFirmID*	–
	16	*WaiverType*	•
	32	*CrossExclusionIndicator*	–
	64	*PriceFormation*	•
	128	*ClientQualifiedRole*	–
11	1	*ClientID*	–
	2	*InvestorID*	–
	4	*ExecutorID*	–
	8	*OrderOrigination*	–
	16	*AlgorithmicIndicator*	•
	32	*DeferralReason*	•
	64	*InvestorQualifiedRole*	–
	128	*ExecutorQualifiedRole*	–
12	1	*CtiCode*	–
	2	*ManualOrderIndicator*	–
	4	*OperatorId*	–
	8	*TradeDate*	–
	16	*VariancePrice*	–
	32	*VarianceSize*	–
	64	*OrigSymbolID*	–
	128	*OrigTASPrice*	–

continued. . .

Byte	Bit	Field	
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	–
	128	*MultilegReportingType*	–
14	1	*LegCFICode*	–
	2	*LegMaturityDate*	–
	4	*LegStrikePrice*	–
	8	*(Reserved)*	–
	16	*SecondaryExecId*	–
	32	*(Reserved)*	–
	64	*(Reserved)*	–
	128	*(Reserved)*	–

6.4 Trade Capture Report Decline V2

Byte	Bit	Field	
1	1	*Side*	•
	2	*PegDifference*	–
	4	*Price*	–
	8	*ExecInst*	•
	16	*OrdType*	–
	32	*TimeInForce*	–
	64	*MinQty*	–
	128	*MaxRemovePct*	–
2	1	*Symbol*	•
	2	*SymbolSfx*	–
	4	*Currency*	•
	8	*IdSource*	•
	16	*SecurityId*	•
	32	*SecurityExchange*	•
	64	*Capacity*	•
	128	*(Reserved)*	–
3	1	*Account*	•
	2	*ClearingFirm*	•
	4	*ClearingAccount*	–
	8	*DisplayIndicator*	–
	16	*MaxFloor*	–
	32	*DiscretionAmount*	–
	64	*OrderQty*	•
	128	*PreventParticipantMatch*	–
4	1	*MaturityDate*	–
	2	*StrikePrice*	–
	4	*PutOrCall*	–
	8	*OpenClose*	–
	16	*ClOrdIdBatch*	–
	32	*CorrectedSize*	–
	64	*PartyID*	•
	128	*AccessFee*	–
5	1	*OrigClOrdId*	–
	2	*LeavesQty*	–
	4	*LastShares*	–
	8	*LastPrice*	–
	16	*DisplayPrice*	–
	32	*WorkingPrice*	–
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	–
6	1	*SecondaryOrderId*	–
	2	*CCP*	–
	4	*ContraCapacity*	–
	8	*AttributedQuote*	–
	16	*ExtExecInst*	–
	32	*BulkOrderIds*	–
	64	*BulkRejectReasons*	–
	128	*PartyRole*	•

continued. . .

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	•
	4	*TradePublishIndReturn*	•
	8	*Text*	•
	16	*Bid*	–
	32	*Offer*	–
	64	*LargeSize*	•
	128	*LastMkt*	–
8	1	*FeeCode*	–
	2	*EchoText*	•
	4	*StopPx*	–
	8	*RoutingInst*	–
	16	*RoutStrategy*	–
	32	*RouteDeliveryMethod*	–
	64	*ExDestination*	–
	128	*TradeReportRefID*	–
9	1	*MarketingFeeCode*	–
	2	*TargetPartyID*	–
	4	*AuctionId*	–
	8	*OrderCategory*	–
	16	*LiquidityProvision*	–
	32	*CmtaNumber*	–
	64	*CrossType*	–
	128	*CrossPrioritization*	–
10	1	*CrossId*	–
	2	*AllocQty*	–
	4	*GiveUpFirmID*	–
	8	*RoutingFirmID*	–
	16	*WaiverType*	•
	32	*CrossExclusionIndicator*	–
	64	*PriceFormation*	•
	128	*ClientQualifiedRole*	–
11	1	*ClientID*	–
	2	*InvestorID*	–
	4	*ExecutorID*	–
	8	*OrderOrigination*	–
	16	*AlgorithmicIndicator*	•
	32	*DeferralReason*	•
	64	*InvestorQualifiedRole*	–
	128	*ExecutorQualifiedRole*	–
12	1	*CtiCode*	–
	2	*ManualOrderIndicator*	–
	4	*OperatorId*	–
	8	*TradeDate*	–
	16	*VariancePrice*	–
	32	*VarianceSize*	–
	64	*OrigSymbolID*	–
	128	*OrigTASPrice*	–

continued. . .

Byte	Bit	Field	
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	–
	128	*MultilegReportingType*	–
14	1	*LegCFICode*	–
	2	*LegMaturityDate*	–
	4	*LegStrikePrice*	–
	8	*(Reserved)*	–
	16	*SecondaryExecId*	–
	32	*(Reserved)*	–
	64	*(Reserved)*	–
	128	*(Reserved)*	–

6.5 Quote Status V2

Byte	Bit	Field	
1	1	*Side*	–
	2	*PegDifference*	–
	4	*Price*	–
	8	*ExecInst*	–
	16	*OrdType*	–
	32	*TimeInForce*	–
	64	*MinQty*	–
	128	*MaxRemovePct*	–
2	1	*Symbol*	•
	2	*SymbolSfx*	–
	4	*Currency*	•
	8	*IdSource*	•
	16	*SecurityId*	•
	32	*SecurityExchange*	•
	64	*Capacity*	•
	128	*(Reserved)*	–
3	1	*Account*	–
	2	*ClearingFirm*	•
	4	*ClearingAccount*	–
	8	*DisplayIndicator*	–
	16	*MaxFloor*	–
	32	*DiscretionAmount*	–
	64	*OrderQty*	–
	128	*PreventParticipantMatch*	–
4	1	*MaturityDate*	–
	2	*StrikePrice*	–
	4	*PutOrCall*	–
	8	*OpenClose*	–
	16	*ClOrdIdBatch*	–
	32	*CorrectedSize*	–
	64	*PartyID*	–
	128	*AccessFee*	–
5	1	*OrigClOrdId*	–
	2	*LeavesQty*	–
	4	*LastShares*	–
	8	*LastPrice*	–
	16	*DisplayPrice*	–
	32	*WorkingPrice*	–
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	–
6	1	*SecondaryOrderId*	–
	2	*CCP*	–
	4	*ContraCapacity*	–
	8	*AttributedQuote*	–
	16	*ExtExecInst*	–
	32	*BulkOrderIds*	–
	64	*BulkRejectReasons*	–
	128	*PartyRole*	–

continued. . .

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	–
	4	*TradePublishIndReturn*	–
	8	*Text*	–
	16	*Bid*	•
	32	*Offer*	•
	64	*LargeSize*	–
	128	*LastMkt*	–
8	1	*FeeCode*	–
	2	*EchoText*	–
	4	*StopPx*	–
	8	*RoutingInst*	–
	16	*RoutStrategy*	–
	32	*RouteDeliveryMethod*	–
	64	*ExDestination*	–
	128	*TradeReportRefID*	–
9	1	*MarketingFeeCode*	–
	2	*TargetPartyID*	–
	4	*AuctionId*	–
	8	*OrderCategory*	–
	16	*LiquidityProvision*	–
	32	*CmtaNumber*	–
	64	*CrossType*	–
	128	*CrossPrioritization*	–
10	1	*CrossId*	–
	2	*AllocQty*	–
	4	*GiveUpFirmID*	–
	8	*RoutingFirmID*	–
	16	*WaiverType*	–
	32	*CrossExclusionIndicator*	–
	64	*PriceFormation*	–
	128	*ClientQualifiedRole*	–
11	1	*ClientID*	–
	2	*InvestorID*	–
	4	*ExecutorID*	–
	8	*OrderOrigination*	–
	16	*AlgorithmicIndicator*	–
	32	*DeferralReason*	–
	64	*InvestorQualifiedRole*	–
	128	*ExecutorQualifiedRole*	–
12	1	*CtiCode*	–
	2	*ManualOrderIndicator*	–
	4	*OperatorId*	–
	8	*TradeDate*	–
	16	*VariancePrice*	–
	32	*VarianceSize*	–
	64	*OrigSymbolID*	–
	128	*OrigTASPrice*	–

continued. . .

Byte	Bit	Field	
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	–
	128	*MultilegReportingType*	–
14	1	*LegCFICode*	–
	2	*LegMaturityDate*	–
	4	*LegStrikePrice*	–
	8	*(Reserved)*	–
	16	*SecondaryExecId*	–
	32	*(Reserved)*	–
	64	*(Reserved)*	–
	128	*(Reserved)*	–

6.6 Quote Reject V2

Byte	Bit	Field	
1	1	*Side*	–
	2	*PegDifference*	–
	4	*Price*	–
	8	*ExecInst*	–
	16	*OrdType*	–
	32	*TimeInForce*	–
	64	*MinQty*	–
	128	*MaxRemovePct*	–
2	1	*Symbol*	•
	2	*SymbolSfx*	–
	4	*Currency*	•
	8	*IdSource*	•
	16	*SecurityId*	•
	32	*SecurityExchange*	•
	64	*Capacity*	•
	128	*(Reserved)*	–
3	1	*Account*	–
	2	*ClearingFirm*	•
	4	*ClearingAccount*	–
	8	*DisplayIndicator*	–
	16	*MaxFloor*	–
	32	*DiscretionAmount*	–
	64	*OrderQty*	–
	128	*PreventParticipantMatch*	–
4	1	*MaturityDate*	–
	2	*StrikePrice*	–
	4	*PutOrCall*	–
	8	*OpenClose*	–
	16	*ClOrdIdBatch*	–
	32	*CorrectedSize*	–
	64	*PartyID*	–
	128	*AccessFee*	–
5	1	*OrigClOrdId*	–
	2	*LeavesQty*	–
	4	*LastShares*	–
	8	*LastPrice*	–
	16	*DisplayPrice*	–
	32	*WorkingPrice*	–
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	–
6	1	*SecondaryOrderId*	–
	2	*CCP*	–
	4	*ContraCapacity*	–
	8	*AttributedQuote*	–
	16	*ExtExecInst*	–
	32	*BulkOrderIds*	–
	64	*BulkRejectReasons*	–
	128	*PartyRole*	–

continued. . .

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	–
	4	*TradePublishIndReturn*	–
	8	*Text*	–
	16	*Bid*	•
	32	*Offer*	•
	64	*LargeSize*	–
	128	*LastMkt*	–
8	1	*FeeCode*	–
	2	*EchoText*	–
	4	*StopPx*	–
	8	*RoutingInst*	–
	16	*RoutStrategy*	–
	32	*RouteDeliveryMethod*	–
	64	*ExDestination*	–
	128	*TradeReportRefID*	–
9	1	*MarketingFeeCode*	–
	2	*TargetPartyID*	–
	4	*AuctionId*	–
	8	*OrderCategory*	–
	16	*LiquidityProvision*	–
	32	*CmtaNumber*	–
	64	*CrossType*	–
	128	*CrossPrioritization*	–
10	1	*CrossId*	–
	2	*AllocQty*	–
	4	*GiveUpFirmID*	–
	8	*RoutingFirmID*	–
	16	*WaiverType*	–
	32	*CrossExclusionIndicator*	–
	64	*PriceFormation*	–
	128	*ClientQualifiedRole*	–
11	1	*ClientID*	–
	2	*InvestorID*	–
	4	*ExecutorID*	–
	8	*OrderOrigination*	–
	16	*AlgorithmicIndicator*	–
	32	*DeferralReason*	–
	64	*InvestorQualifiedRole*	–
	128	*ExecutorQualifiedRole*	–
12	1	*CtiCode*	–
	2	*ManualOrderIndicator*	–
	4	*OperatorId*	–
	8	*TradeDate*	–
	16	*VariancePrice*	–
	32	*VarianceSize*	–
	64	*OrigSymbolID*	–
	128	*OrigTASPrice*	–

continued. . .

Byte	Bit	Field	
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	–
	128	*MultilegReportingType*	–
14	1	*LegCFICode*	–
	2	*LegMaturityDate*	–
	4	*LegStrikePrice*	–
	8	*(Reserved)*	–
	16	*SecondaryExecId*	–
	32	*(Reserved)*	–
	64	*(Reserved)*	–
	128	*(Reserved)*	–

7 List of Optional Fields

The following are descriptions of optional fields which may be sent or received.

Field	Length	Data Type	Description
Account	16	Text	Corresponds to *Account* (1) in Bats FIX. Reflected back on execution reports associated with this order. May be made available in the Participant's clearing file. Allowed characters are alphanumeric and colon.
Algorithmic Indicator	1	Text	This corresponds to *AlgorithmicTradeIndicator* (2667) in Bats FIX. Indicates the Trade Capture Report was placed as a result of an investment firm engaging in algorithmic trading. `N` = No algorithm was involved (default). `Y` = Algorithm was involved (ALGO).
BidPx	8	Binary Price	Corresponds to *BidPx* (132) in FIX. Specifies the bid quote price. If omitted, indicates the bid quote is not to be changed, and BidSize must also be omitted. If BidPx = 0, indicates the bid quote is to be cancelled, and BidSize = 0 must also be present
BidSize	4	Binary	Corresponds to *BidSize* (132) in FIX. The quantity of the bid quote. Must be zero when cancelling a quote.
CancelOrig OnReject	1	Alpha	Corresponds to *CancelOrigOnReject* (9619) in Bats FIX. Indicates handling of original order on failure to modify. `N` = Leave original order alone. `Y` = Cancel original order if modification fails.
Capacity	1	Alpha	Corresponds to *OrderCapacity* (47) in Bats FIX. (Orders). Corresponds to *LastCapacity* (29) in Bats FIX. (Executions). `A` = Agency (maps to 'AOTC') `P` = Principal (maps to 'DEAL') `R` = Riskless Principal (maps to 'MTCH')
Central Counterparty	1	Alpha	The CCP handling the trade `N` = None
ClearingFirm	4	Alpha	Corresponds to *OnBehalfOfCompID* (115) and *ClearingFirm* (439) in Bats FIX. Firm that will clear the trade. If empty (all binary zero), a default will be used (only permitted on non-service bureau accounts).
CorrectedSize	4	Binary	Corresponds to *CorrectedSize* (6655) in Bats FIX. Number of shares after trade adjustment.
Currency	3	Alpha	Corresponds to *Currency* (15) in Bats FIX. ISO currency. Required if *IDSource* is set to 4 (ISIN).

DeferralReason	1	Alphanumeric	Corresponds to *TrdRegPublicationReasons* (8013) in FIX and is used to indicate the deferral reason used for the reported trade. The following values are valid: - = No Deferral Reason 6 = Deferral for Large In Scale (LRGS) ~~(Bats to Participant only)~~ 7 = Deferral for Illiquid Instrument (for RTS2 only) (ILQD) 8 = Deferral for Size Specific (for RTS2 only) (SIZE) C = Deferral for Illiquid Instrument (for RTS2 only) (ILQD) **AND** Deferral for Large In Scale (LRGS) D = Deferral for Illiquid Instrument (for RTS2 only) (ILQD) **AND** Deferral for Size Specific (for RTS2 only) (SIZE) Each Deferral Reason is represented by a distinct value that can be requested individually. Deferrals that can be applied together are represented by values that indicates the deferrals being set together. Based upon what is requested, the system calculates which of these are valid. The business confirmation contains the deferral(s) that have been appplied.
ExecInst	1	Text	Corresponds to *ExecInst* (18) in Bats FIX. P = Market Peg (peg buy to PBBO offer, peg sell to PBBO bid) R = Primary Peg (peg buy to PBBO bid, peg sell to PBBO offer) M = Midpoint (peg to PBBO midpoint) L = Alternate Midpoint (less aggressive of midpoint and 1 tick inside PBBO) **for Periodic Auction Orders:**[3] M = Midpoint (peg to Bats EBBO midpoint) G = Guarded Midpoint (peg to Bats EBBO midpoint but suspend order if primary market quote becomes one-sided or disappears) **for Smart Order Routing:** u = Bats + External Dark Only v = Bats + External Dark + Lit w = Bats + External Lit Only ASCII NUL (0x00) = no special handling Default = ASCII NUL (0x00)
ExecutionMethod	1	Alpha	Corresponds to *ExecutionMethod* (2405) in FIX. Optional. Is used by the participant to indicate the method by which the trade was executed. This field corresponds to the proposed MMT Level 3.7 (Offbook Automated Liquidity Indicator). The following values are valid: A = Automated M = Manual U = Unspecified (default)

[3] RoutingInst=BP

ExpireTime	8	DateTime	Corresponds to *ExpireTime* (126) in FIX. Required for *TimeInForce* = 6 orders, specifies the date-time (in UTC) that the order expires.
GrossTradeAmt	8	Binary Price	Total amount traded, expressed in units of currency. A value of zero or an indicative price may be used if the price is pending, as denoted by *PriceFormation*
IDSource	1	Alphanumeric	Corresponds to *IDSource* (22) in Bats FIX. 4 = ISIN 5 = RIC
LargeSize	8	Binary	Number of shares relevant for the trade. Used when size exceeds the capabilities of 32-bit. System limit is 99,999,999,999.
LastMkt	4	Alphanumeric	Corresponds to *LastMkt* (30) in Bats FIX. Segment MIC of this fill.
LastPx	8	Binary Price	Corresponds to *LastPx* (31) in Bats FIX. Price of this fill. A value of zero or an indicative price may be used if the price is pending, as denoted by *PriceFormation*
LastShares	4	Binary	Corresponds to *LastShares* (32) in Bats FIX. Executed share quantity. If the LargeSize optional field is specified, that value holds precedance over this field.
MatchType	1	Binary	Corresponds to *MatchType* (574) in FIX. The following values are valid: 1 = Trade Reporting (Off-Exchange) 3 = Trade Reporting (On-Exchange) 9 = Trade Reporting (Systematic Internalizer)
OfferPx	8	Binary Price	Corresponds to *OfferPx* (133) in FIX. Specifies the offer quote price. If omitted, indicates the offer quote is not to be changed, and OfferSize must also be omitted. If OfferPx = 0, indicates the offer quote is to be cancelled, and OfferSize = 0 must also be present.
OfferSize	4	Binary	Corresponds to *OfferSize* (132) in FIX. The quantity of the offer quote. Must be zero when cancelling a quote.
OrderCategory	1	Binary	Optional. This field corresponds to the MMT Level 3.2 field 'Negotiated Transaction Indicator', and is used by the participant to indicate that the trade made use of the Negotiated Transaction MiFID waiver. 0 = Not a Negotiated Trade 3 = Privately Negotiated Trade On return fields, this field indicates whether Bats deems the trade as utilising the Negotiated Transaction waiver under MiFID.

PriceFormation	1	Alphanumeric	Optional. Indicates the price formation attribute of the trade, and corresponds to MMT v3 Level 3.8 Not specified or P = Plain-Vanilla Trade J = Trade Not Contributing to the Price Discovery Process (TNCP) N = Price is Pending (PNDG)
Price	8	Binary Price	Corresponds to *Price* (44) in Bats FIX. Limit price. Four implied decimal places.
RptTime	8	DateTime	Corresponds to *RptTime* (7570) in FIX. Optional. Indicates the time at which a deferred trade report will be automatically published.
SecondaryOrderID	8	Binary	Corresponds to *SecondaryOrderID* (198) in Bats FIX. Denotes an alternative *OrderID* which is present on Bats market data feeds (for example, to hide that a reserve (iceberg) order has reloaded). Or, *OrderID* of the contra side of a prevented match.
Side	1	Alphanumeric	Corresponds to *Side* (54) in Bats FIX. 1 = Buy 2 = Sell 8 = Cross
SecondaryTrdType	1	Binary	Corresponds to *SecondaryTrdType* (855) in FIX. The following values are valid: 64 = Benchmark Trade
SecurityExchange	4	Alphanumeric	Corresponds to *SecurityExchange* (207) in Bats FIX. Required if *IDSource* is set to 4 (ISIN).
SecurityID	16	Text	Corresponds to *SecurityID* (48) in Bats FIX. ISIN, or RIC if *IDSource* is set.
SettlementDate	8	DateTime	Used to specify the date on which the trade is desired to settle. Note, the actual settlement date may be varied by the central counterparties (CCPs) due to operational requirements (eg. for symbols in a conditional trading status). May only be specified on a new trade report.
SettlementPrice	8	Trade Price	Price at which the trade should settle at. If specified, any risk controls will be applied against this price.
Symbol	8	Alphanumeric	Corresponds to *Symbol* (55) in Bats FIX. Uniform symbology identifier for the instrument.
Text	60	Text	Used to provide additional textual information, e.g. a warning.

TimeInForce	1	Alphanumeric	Corresponds to *TimeInForce* (59) in FIX. 0 = Day 1 = GTC (allowed, but treated as Day) 2 = At The Open 3 = IOC (Portion not filled immediately is cancelled. Market orders are implicitly IOC.) 6 = GTD (expires at earlier of specified *ExpireTime* or end of day) 7 = At The Close 8 = Good For Auction (only valid if *RoutingInst*=BP)
TradeID	8	Binary	Corresponds to *TradeID* (1003) in FIX. Optional. Is used by the participant to specify the previously reported trade that the report sent refers to.
TradePrice Condition	1	Binary	Corresponds to *TradePriceCondition* (1390) in FIX. The following values are valid: 0 = Cum Dividend (deprecated) 2 = Ex Dividend (deprecated) 13 = Special Dividend (SDIV)
TradePublish Indicator	1	Binary	Corresponds to *TradePublishIndicator* (1390) in FIX. Optional. Is used by the participant to request that the publication be delayed. The following values are valid: 0 = Do not publish 1 = Publish trade 2 = Deferred publication
TradeReport TransType	1	Binary	Corresponds to *TradeReportTransType* (487) in FIX. Optional. Specifies the transaction type of the report sent via Trade Capture Report. The following values are valid: 0 = New 1 = Cancel 2 = Replace 3 = Release
TradeTime	8	DateTime	Corresponds to *TransactTime* (60) and *TradeDate* (75) in FIX. Optional, for new trade reports. Cancel/amend/releases require the original time of the trade. Specifies the date and time at which the trade was arranged. This field defaults to the time at which the message is received, when defaulting is allowed.

TradingSession SubId	1	Binary	Corresponds to *TradingSessionSubId* (625) in FIX. The following values are valid: 2 = Scheduled Opening Auction 4 = Scheduled Closing Auction 6 = Scheduled Intraday Auction 8 = Unspecified Auction 9 = Unscheduled Auction 3 = Continuous Trading 5 = Post Trading 10 = Out of Main Session Trading
Transaction Category	1	Alphanumeric	Corresponds to *TrdType* (828) in FIX. Also corresponds to *Transaction Category* in MMT. Optional. Specifies the type or category of the trade being reported in a Trade Capture Report. At this time, only the following values are valid: P = Regular Trade (aka Plain-Vanilla Trade) D = Dark Trade R = Trade With Price Improvement (RPRI)
TrdSubType	1	Binary	Corresponds to *TrdSubType* (829) in FIX. Optional. The following values are valid: 37 = Agency Cross trade
VenueType	1	Alphanumeric	Corresponds to *VenueType* (1430) in FIX. The following values are valid: O = Off Book B = Central Limit Order Book Q = Quote Driven Market D = Dark Order Book A = Periodic Auction N = Request For Quotes H = Hybrid Market

WaiverType	1	Alphanumeric	Corresponds to *TrdRegPublicationReasons* (8013) in FIX and is used to indicate the waiver used for the reported trade. The following values are valid: - = No Waiver Type 3 = Reference Price (Dark Book) (for MTF only) (RFPT) 4 = Pre-Trade Transparency Waiver for Illiquid Instrument (for SI only) (ILQD) 5 = Pre-Trade Transparency Waiver for Above Standard Market Size (for SI only) (SIZE) B = Pre-Trade Transparency Waiver for Illiquid Instrument (for SI only) (ILQD) **AND** Pre-Trade Transparency Waiver for Above Standard Market Size (for SI only) (SIZE) Each Pre-Trade Transparency Waiver is represented by a distinct value that can be requested individually. Waivers that can be applied together are represented by a value that indicates both waivers being set together. Based upon what is requested, the system calculates which of these are valid. The business confirmation contains the waiver(s) that have been appplied.

8 Reason Codes

The following is a list of all reason codes used. These reason codes are used in a variety of contexts (order cancellations, order rejections, modify rejections, etc.). All reasons are not valid in all contexts. Bats may add additional reason codes without notice. Members must gracefully ignore unknown values.

- `A` = Admin
- `D` = Duplicate Identifier (e.g., *TradeReportID*)
- `Y` = Symbol Not Supported
- `Z` = Unforeseen Reason
- `m` = Market Access Risk Limit Exceeded

9 List of Message Types

9.1 Participant to Bats

Message Name	Level	Type	Sequenced
Login Request V2	Session	0x37	No
Logout Request	Session	0x02	No
Client Heartbeat	Session	0x03	No
Trade Capture Report V2	Application	0x3C	Yes
Quote V2	Application	0x3D	Yes
Quote Cancel V2	Application	0x3E	Yes

9.2 Bats to Participant

Message Name	Level	Type	Sequenced
Login Response V2	Session	0x24	No
Logout	Session	0x08	No
Server Heartbeat	Session	0x09	No
Replay Complete	Session	0x13	No
Trade Capture Report Accept V2	Application	0x30	Yes
Trade Capture Report Reject V2	Application	0x31	No
Trade Capture Report Confirm V2	Application	0x32	Yes
Trade Capture Report Decline V2	Application	0x33	Yes
Quote Status V2	Application	0x3F	Yes
Quote Reject V2	Application	0x40	Yes

10 Port Attributes

The table below lists BOE port attributes that are configurable on the port or firm level. Changes to these attributes can be made by contacting the Bats Trade Desk.

Attribute	Default	Description
Allowed Clearing Executing Firm ID(s)	No MPIDs	Executing Firm ID(s) allowed for quoting on the port.
Allowed Trade Reporting Firm ID(s)	No MPIDs	Executing Firm ID(s) allowed for trade reporting on the port.
Cancel on Disconnect	Option 1	Bats offers two options for cancelling orders as a result of a session disconnect: 1. Cancel all open orders (continuous book and on-open, on-close and periodic auction orders). 2. Do not cancel any open orders.
Send Trade Breaks^	No	Enables sending of TRADE CANCEL OR CORRECT V2 messages.
Reject Orders on DROP Port Disconnect	No	Allows Participant/Sponsoring Firms to associate DROP port(s) to order entry port(s). If all associated DROP ports experience disconnection, new orders will be rejected until at least one DROP port session has been reestablished.
Reject Orders on DROP Port Disconnect	30 seconds	Only applicable if "Reject Orders on DROP Port Disconnect" has been enabled. When the last associated DROP port has disconnected, begin rejecting orders on the associated order entry port(s) if a DROP session has not been reestablished within this timeout. Minimum value allowed is 0 seconds.
Cancel Open Orders on DROP Port Disconnect	No	Only applicable if "Reject Orders on DROP Port Disconnect" has been enabled. When the last associated DROP port has disconnected, cancel all associated open orders.

†Port attribute can be overridden on an order-by-order basis.

^Requires certification.

11 Support

Please email questions or comments regarding this specification to `tradedeskeurope@bats.com`.

Revision History

July 19, 2017	Version 2.0.17 MMT v3.04 support for Q4 2017 release.
July 10, 2017	Version 2.0.16 Added *FeeCode* to repeating group on TRADE CAPTURE CONFIRM V2 messages.
June 2, 2017	Version 2.0.15 Clarified *TransactionCategory*=R maps to MMT v3 'RPRI' on market data.
May 24, 2017	Version 2.0.14 Corrected description for *PriceFormation*. Driving MMT v3 'NPFT' on market data not a valid option off-exchange.
March 2, 2017	Version 2.0.13 Add new field type *Date*
February 9, 2017	Version 2.0.12 Review feedback for MMT v3
February 1, 2017	Version 2.0.11 Support for MMT v3
December 9, 2016	Version 2.0.10 Update fields on QUOTE STATUS V2 and QUOTE REJECT V2
August 4, 2016	Version 2.0.9 Update field length for GrossTradeAmt
June 15, 2016	Version 2.0.8 Fix typo regarding size of PartyRole field, plus its omission from return TradeCapture messages
February 19, 2016	Version 2.0.7 Updated for new branding.
January 8, 2016	Version 2.0.6 Added Order Category optional return bitfield.
December 1, 2015	Version 2.0.5 For TRADE CAPTURE REPORT V2, clarified that TradeTime is only optional for new trades.
June 13, 2015	Version 2.0.4 Added 5th input bitfield for TRADE CAPTURE REPORT V2.
May 28, 2015	Version 2.0.3 Added an example for TRADE CAPTURE REPORT V2 and clarified description of *TrdCapRptSideGrp*.
March 13, 2015	Version 2.0.2 Added *TradeReportRefID* to ROB8.
March 12, 2015	Version 2.0.1 Added detail on fields in TCRB4.
February 10, 2015	Version 2.0 First Version 2 release.